

JEE MAIN REVISION MASTER: PROBABILITY

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 44 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Conditional Probability, Multiplication Rule & Independence	28 PYQs	Conditional Probability
Very High	Bayes' Theorem & Binomial Distribution	16 PYQs	Distributions & Inference

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Conditional Probability, Multiplication Rule & Independence	$P(A B) = \frac{P(A \cap B)}{P(B)}$, $\text{Independent: } P(A \cap B) = P(A) \cdot P(B)$, $P(A' \cap B') = [1 - P(A)][1 - P(B)]$
Bayes' Theorem & Binomial Distribution	$P(X=k) = {}^nC_k p^k q^{n-k}$, $\text{Mean} = np$, $\text{Variance} = npq$, $\text{Variance} < \text{Mean}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Conditional Probability, Multiplication Rule & Independence (Very High Yield — 28 PYQs)

Core Principle: Conditional probability $P(A|B) = (P(A \cap B) / P(B))$ ($P(B) > 0$). Independent events: Occurrence of A does not affect B; condition for independence is $P(A \cap B) = P(A) \cdot P(B)$. Mutually exclusive events: cannot occur simultaneously $\implies P(A \cap B) = 0$.

■ **Caution / JEE Pitfall:** Mutually exclusive events with non-zero probabilities can NEVER be independent because $P(A \cap B) = 0 \neq P(A)P(B)$!

■ **Speed Trick:** If A and B are independent, then A' and B' are also independent: $P(A' \cap B') = P(A') P(B') = (1-P(A))(1-P(B))$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "Probability of at least one of the events A and B occur is 0.6 . If A and B occur simultaneously with probability 0.2 , then $P(\bar{A})$..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "The probability that a man and his wife live after 20 years are (1 / 4) and (1 / 3) respectively. The probability that neither..." [Ans: D]
- [2026 Shift] Q#3 (MCQ): "Probability of obtaining an even prime number on each die when a pair of dice is rolled is..." [Ans: D]

3.2 Bayes' Theorem & Binomial Distribution (Very High Yield — 16 PYQs)

Core Principle: Bayes' Theorem for posterior probability: $P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum P(E_j)P(A|E_j)}$. Binomial distribution $B(n, p)$: $P(X = k) = {}^nC_k p^k q^{n-k}$ ($q = 1-p$). Mean $\mu = np$; Variance $\sigma^2 = npq$; Standard deviation $\sigma = \sqrt{npq}$. Since $q < 1$, the variance of a binomial distribution is ALWAYS strictly less than its mean ($\sigma^2 < \mu$).

■ **Caution / JEE Pitfall:** In binomial distribution, variance can NEVER exceed the mean ($npq < np$ always because $q < 1$). If given a problem where variance $>$ mean, that distribution is impossible!

■ **Speed Trick:** Probability of at least one success in n trials: $P(X \geq 1) = 1 - P(X = 0) = 1 - q^n$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "Recent studies suggest that 12% of the world population is left handed. Depending on parents' hand usage, the chances of having left handed..." **[Ans: N/A]**
- **[2024 Shift] Q#1 (MCQ):** "A random variable X has the following probability distribution: X 0 1 2 $P(X)$ $\frac{25}{36}$ $\frac{k}{36}$ If the mean of the random variable X is ..." **[Ans: N/A]**
- **[2024 Shift] Q#2 (MCQ):** "If a random variable X follows the binomial distribution with parameters $n=5, p$ and $P(X=2)=9P(X=3)$, then p is equal to..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

Recent studies suggest that 12% of the world population is left handed. Depending on parents' hand usage, the chances of having left handed children are as follows: A: Both parents are left handed, chances of having left handed children =24% B: Both parents are right handed, chances of having left handed children =9% C: Father left handed and mother right handed, chances of having left handed children =17% D: Father right handed and mother left handed, chances of having left handed children =22% Given $P(A)=P(B)=P(C)=P(D)=1/4$ and L denotes child is left handed. What is the probability that $P(A|L)$?

- (A) $(17 / 100)$
- (B) $(19 / 25)$
- (C) $(1 / 3)$
- (D) $(2 / 3)$

Correct Answer: None

Solution Analysis: We need to find $P(A|L)$, which means: Probability that both parents are left handed, given that the child is left handed. Using Bayes' theorem, $P(A|L) = \frac{P(A)P(L|A)}{P(L)}$ Here, $P(A)=P(B)=P(C)=P(D)=(1/4)$ Also, $P(L|A)=24\%=(24/100)$ Now find $P(L)$ using the total probability formula:

$P(L) = P(A)P(L|A) + P(B)P(L|B) + P(C)P(L|C) + P(D)P(L|D)$ Substitute the values: $P(L) = (1/4) \times (24/100) + (1/4) \times (9/100) + (1/4) \times (17/100) + (1/4) \times (22/100)$ Since $(1/4)$ is common, $P(L) = (1/4)((24+9+17+22/100))$ $P(L)$

Benchmark Problem 2: KCET 2026 (Q#1)

Probability of at least one of the events A and B occur is 0.6 . If A and B occur simultaneously with probability 0.2 , then $P(\bar{A})+P(\bar{B})$ is

- (A) 1
- (B) 0.8
- (C) 0.6
- (D) 1.2

Correct Answer: None

Solution Analysis: Given: $P(A \cup B) = 0.6$ and $P(A \cap B) = 0.2$ We use the formula: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ So, $0.6 = P(A) + P(B) - 0.2$ $P(A) + P(B) = 0.8$ Now, $P(\bar{A}) = 1 - P(A)$, $P(\bar{B}) = 1 - P(B)$ So, $P(\bar{A}) + P(\bar{B}) = (1 - P(A)) + (1 - P(B)) = 2 - (P(A) + P(B)) = 2 - 0.8 = 1.2$ So the correct answer is: 1.2 Option D